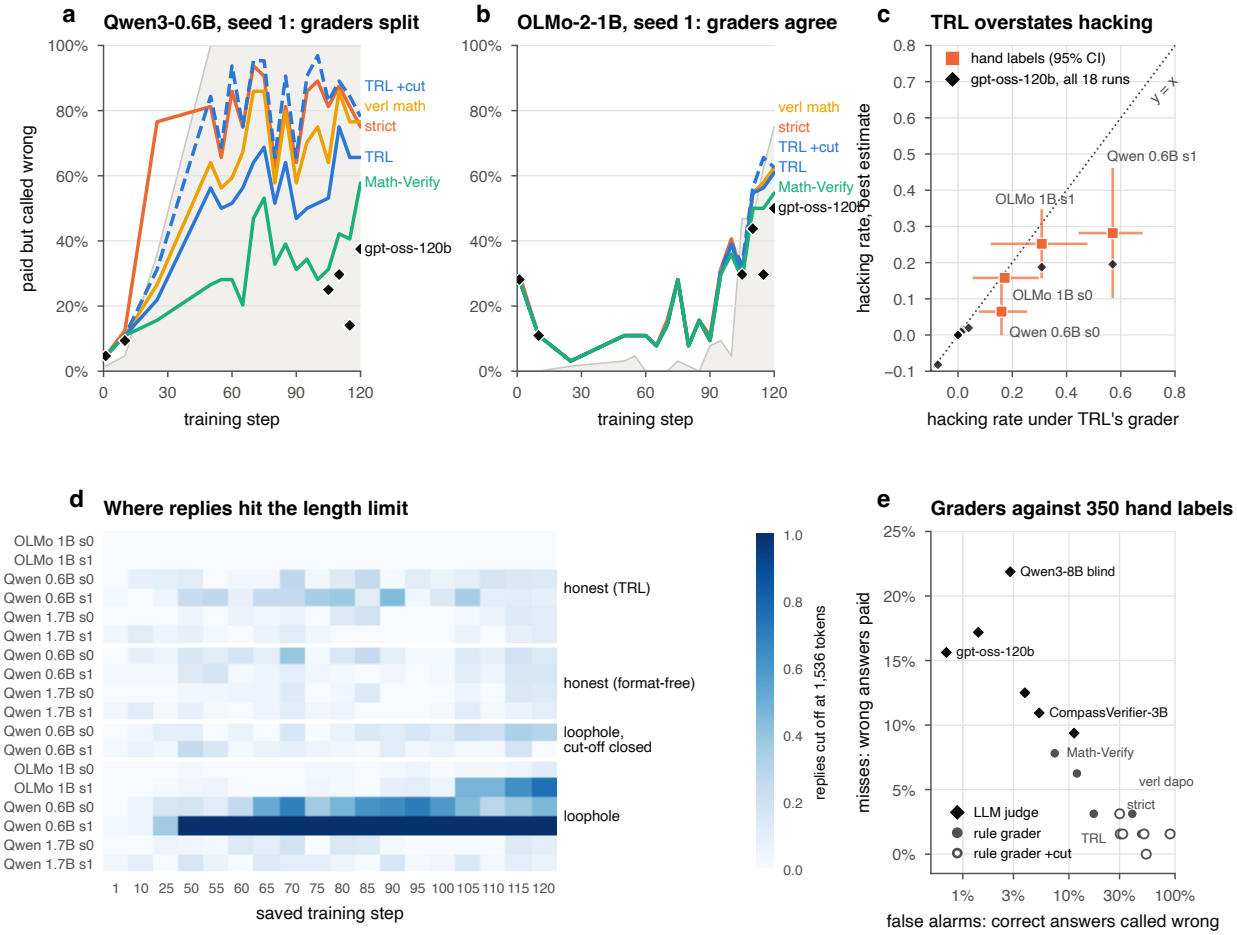
a
Qwen3-0.6B, seed 1: graders split
paid but called wrong
training step
TRL +cut
verl math
strict
TRL
Math-Verify
gpt-oss-120b
b
OLMo-2-1B, seed 1: graders agree
training step
verl math
strict
TRL +cut
TRL
Math-Verify
gpt-oss-120b
c
TRL overstates hacking
hand labels (95% CI)
gpt-oss-120b, all 18 runs
y = x
Qwen 0.6B s1
OLMo 1B s1
OLMo 1B s0
Qwen 0.6B s0
hacking rate, best estimate
hacking rate under TRL's grader
d
Where replies hit the length limit
OLMo 1B s0
OLMo 1B s1
Qwen 0.6B s0
Qwen 0.6B s1
Qwen 1.7B s0
Qwen 1.7B s1
honest (TRL)
honest (format-free)
loophole, cut-off closed
loophole
saved training step
replies cut off at 1,536 tokens
e
Graders against 350 hand labels
Qwen3-8B blind
gpt-oss-120b
CompassVerifier-3B
Math-Verify
verl dapo
strict
TRL
LLM judge
rule grader
rule grader +cut
misses: wrong answers paid
false alarms: correct answers called wrong